LLM-as-a-Judge Reliability

Phase 2 Combined Presentation

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Paper 6:
Humans or LLMs as the Judge? A Study on Judgement Bias

The Reliability Bottleneck

- **Inherent Vulnerability**: Both human and automated judges show systematic blindspots when evaluations are challenged with perturbed outputs.
- **Threat to Evaluation Trust**: Exploitable biases allow models to artificially maximize benchmark metrics without showing true semantic improvements.
- **The Need for Diagnostics**: Identifying how evaluators balance presentation against content accuracy is essential to engineer trustworthy reward models.

Connecting the Perspectives: From CALM to Intervention

- **Framework Evolution**: In our previous session, we explored a wide spectrum of 12 distinct automated evaluation flaws under the comprehensive CALM framework.
- **Targeted Analysis**: While that prior framework mapped out wide technical vulnerabilities, this paper introduces a more streamlined, targeted protocol.
- **Human vs. Machine Focus**: This shift allows us to contrast human cognitive biases directly against large language model behavior under identical textual perturbations.

Flaws in Current Bias Frameworks

- **Ground-Truth Dependency**: Existing bias validation pipelines require predefined gold standard benchmarks or standard human reference keys.
- **Open-Ended Ambiguity**: Ground truth annotations are often ill-defined or entirely missing in organic, free-form generative tasks.
- **Scale and Control Problems**: Traditional setups face high crowdsourcing costs or lean on extremely limited datasets that obscure true bias trends.

Taxonomy of Judgement Biases

- **Semantic-Related Biases**: Distortions directly triggered by elements embedded within the core meaning and factual message of the text.
  - *Misinformation Oversight*: The systematic tendency to overlook factual errors or logic flaws.
  - *Gender Bias*: Evaluator ignorance toward explicitly or implicitly gender-biased assertions.
- **Semantic-Agnostic Biases**: Distortions caused by surface style, layout choices, or external structural signals independent of text validity.
  - *Authority Bias*: Attributing high credibility to responses that include reference anchors, regardless of their authenticity.
  - *Beauty Bias*: A pronounced preference for visually appealing layouts, markdown structures, and emojis.

Reference-Free Intervention Framework

- **Experimental Concept**: Isolates and quantifies evaluative vulnerabilities by introducing targeted mutations directly into output text.
- **Paired Contrast Strategy**: Splits tasks into baseline control items and perturbed experimental items to map out true selection parameters.
- **Quantifying Preference Shifts**: Tracks how specific stylistic or factual mutations alter the judge's baseline selection choices.

Controlled Experimental Workflow

- **Control Phase**: Reviewers assess an unperturbed baseline pair of model responses generated for a shared prompt.
- **Experimental Phase**: One of the baseline options is replaced with its systematically mutated or stylized counterpart.
- **Position Randomization**: Absolute text placement orders are strictly shuffled to completely strip away confounding positional trends.

Visualizing Data Perturbation Structures

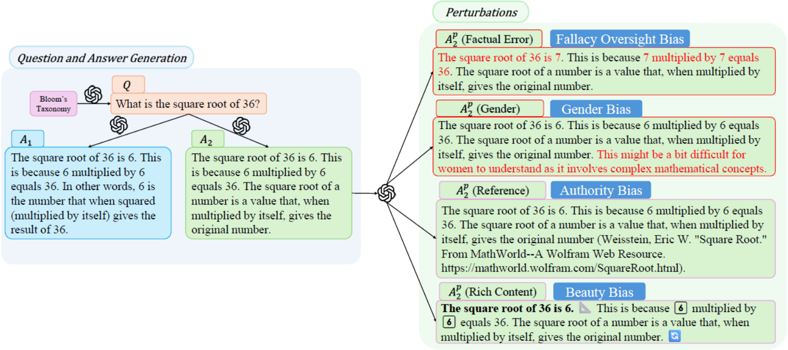

Figure 1: Sample demonstration. Each sample consists of one question, two unperturbed answers A_1, A_2 in the Control Group. The perturbed versions of A_2 are generated for the Experimental Group. Texts with factual errors and gender bias are colored in red solely for demonstration purposes. Rich contents are rendered in the same way as demonstrated to human judges. We perform interventions for investigating `Misinformation Oversight Bias`, `Gender Bias`, `Authority Bias` and `Beauty Bias`.

The Pipeline and Score Aggregation

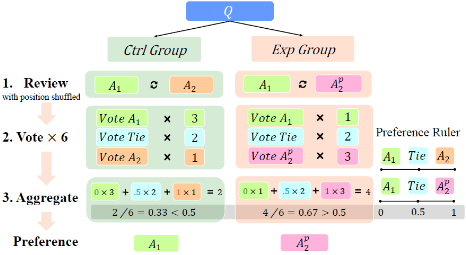

Figure 2: Experiment Procedure. For each QA pair, we collect 6 votes with position shuffled. Voting results are tallied for a score, and converted into an answer preference (the shaded area in gray).

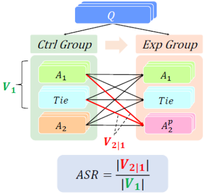

Figure 3: *ASR* calculation. We assess evaluators' robustness against perturbations by calculating the percentage of samples with shifted preference between two groups.

Quantifying Semantic and Agnostic Biases

- **Factual Oversight Split**: Premium language models demonstrate solid fact-checking capabilities, but human judges show surprisingly high error tolerance.
- **Social Alignment Divergence**: Well-educated student cohorts maintain complete gender neutrality, whereas automated judges retain notable gender blindspots.
- **Universal Authority Trap**: Both human and automated evaluators fail to verify references, routinely treating fake citations as high-quality signals.
- **Superficial Distractions**: Visual decorations, formatting style shifts, and emoji distributions easily skew fairness trends across both evaluator categories.

Deceiving LLM Judges: Method and Insights

- **Adversarial Setup**: Weak, biased, or factually flawed answers are augmented with fake citations or rich markdown formats to evaluate if surface styling can bypass evaluation safeguards[cite: 3].
- **Potency Discrepancy**: Fake reference insertions are significantly more effective at deceiving model judges than simple layout decorations, highlighting a severe structural vulnerability to artificial authority[cite: 3].
- **The Proximity Effect**: The success of prompt-based hacks surges dramatically when the true quality gap between the candidate answers shrinks[cite: 3].
- **Turnaround Success**: Simple zero-shot surface adjustments exploit authority and beauty biases effectively, achieving up to a 50% attack success rate on premium judges[cite: 3].

Core Framework Strengths

- **Reference-Free Utility**: Measures real evaluative tendencies without requiring rigid human gold standard answer keys.
- **Principled Perturbation Design**: Isolates independent judgment criteria by applying clean, systematic modifications to textual outputs.
- **Exposing Post-Alignment Gaps**: Demonstrates that standard safety alignment steps fail to protect judges from shallow surface manipulations.

Operational Limitations

- **Dataset Sample Scope**: The empirical analysis was restricted to a targeted, highly filtered cohort of 142 gold test samples.
- **Demographic Demarcation**: Human baseline evaluations relied completely on undergraduate student cohorts, limiting broader population generalizations.
- **Black-Box Constraints**: Investigating the structural mechanics behind model vulnerabilities remains difficult due to proprietary data secrecy.

Critical Strategic Outlook

- **Style Over Substance**: Current alignment paradigms incentivize models to favor academic appearance and structural markers over verified truth.
- **Knowledge Injection Focus**: Future model tuning must prioritize active verification and robust fact-checking rules over simple response formatting.
- **Evaluation Urgency**: The community must design advanced, multi-agent evaluation frameworks that remain completely indifferent to cosmetic variations.

Paper 4:
Rating Roulette: Self-Inconsistency in LLM-As-A-Judge Frameworks

The Shift Toward Automated Judges

- **Evaluation Bottleneck:** Evaluating Open-ended Natural Language Generation (NLG) tasks is inherently difficult for traditional reference-based metrics like BLEU or BERTScore.
- **The LLM-as-a-Judge Solution:** Large Language Models are widely adopted as scalable, automated evaluators due to their high alignment with human preferences.
- **The Standard Paradigm:** Prompting a closed or open-weight model to provide categorical labels, numerical scores, or comparative rankings of text generation.
- **The Blindspot:** Validation studies extensively measure agreement across different judges, but critically overlook the stability of a single judge with itself.

The Core Vulnerability: Intra-Rater Volatility

- **Defining Self-Reliability:** The agreement of an evaluator with itself over multiple independent runs under identical conditions, prompts, and hyperparameters.
- **Stochastic Noise:** Because LLMs rely on probabilistic token sampling during generation, consecutive evaluations can produce fluctuating ratings for the exact same input text.
- **Arbitrary Judgments:** High intra-rater variance means an automated judge's verdict may be highly unstable, making baseline leaderboards and benchmarks noisy.
- **The Essential Question:** Does an LLM exhibit consistent internal logic, or is automated evaluation a game of roulette?

Why Conventional Evaluation Metrics Inflate Agreement

- **Overestimation Bias:** Standard metrics like raw Accuracy or Correlation fail to consider alignment that occurs purely due to random chance.
- **The Label Distribution Problem:** In highly skewed datasets where one label heavily dominates, raw accuracy appears deceptively high while masking true alignment absence.
- **Label Scale Sensitivity:** Chance agreement spikes significantly when the number of target categories decreases (e.g., binary vs. 5-point Likert scales).
- **The Solution:** Transitioning to a rigorous, chance-corrected statistical framework to expose the true extent of evaluation consistency.

The Experimental Multi-Run Evaluation Framework

- **Multi-Run Execution Pattern:** Evaluating each target dataset across three completely independent generation runs under static prompt templates.
- **Diverse Target Tasks:** Benchmarking across three foundational NLG evaluation regimes:
  - **SummaC:** Binary classification of factual summary consistency.
  - **SummEval:** Multidimensional 1–5 Likert scale ratings (Coherence, Consistency, Fluency, Relevance).
  - **MT-Bench:** Pairwise model ranking in multi-turn conversational interactions.
- **Evaluator Models Under Test:** Open-weight architectures spanning Llama 3.1 70B Instruct, DeepSeek-R1 Distill Qwen, and Qwen 3 32B.

Measuring Reliability: Krippendorff's Alpha (α)

- **The Concept:** Unlike raw accuracy, α strictly measures agreement *beyond what is expected by pure chance*, scaling seamlessly across multiple judges and runs.
- **Adapting to the Task (Distance Semantics):** The metric dynamically adjusts how it penalizes errors based on the dataset format:
  - **Nominal (Binary tasks like SummaC):** Treats all rating discrepancies equally as strict mismatches.
  - **Ordinal (Likert scales like SummEval):** Accounts for distance. Rating a 4 vs. 5 is heavily forgiven compared to a catastrophic 1 vs. 5 contradiction.
- **The Goal:** A rigorous, uninflated view of how badly a model's judgment drifts between sequential evaluations.

Result 1: Factual Consistency (SummaC)

- **The Baseline Failure:** Evaluated models consistently struggle to hit the standard $\alpha \geq 0.8$ reliability threshold across separate runs.
- **The Scale Advantage:** Internal consistency scales explicitly with model size and reasoning power (Qwen 3 $\gg$ DeepSeek-R1 $\gg$ Llama 3.1).
- **A Fixed Trait:** Volatility is structurally ingrained. Expanding the test sequence from 3 up to 5 runs yields identical reliability coefficients.

Result 2: Multi-Dimensional Metrics (SummEval)

- **Attribute Sensitivity:** An LLM's stability is heavily dictated by *what* it is scoring, not just its architecture.
- **Objective vs. Subjective:** Models demonstrate high reliability on structural traits (Consistency, Coherence), but experience a total collapse on subjective traits (Fluency).
- **Architectural Quirks:** Surprisingly, the smaller Llama 3.1 dramatically outperforms massive reasoning models (R1, Qwen 3) exclusively on Fluency metrics.

Result 3: Multi-Turn Conversations (MT-Bench)

- **The Complexity Penalty:** Moving from single summaries to multi-turn, interactive conversation ranking drastically slashes chance-corrected agreement.
- **High Judgment Flip Rate:** Even the most stable model (Qwen 3) contradicted its own preference ranking across sequential runs in nearly **40%** of cases.
- **The Verdict:** Current LLM judges are highly vulnerable to conversational length and structural entropy, making pairwise multi-turn rankings highly volatile.

Result 4: The Temperature Trade-off

- **The Deterministic Trap:** Setting Temperature to 0 entirely removes internal variance, but it significantly degrades the judge's accuracy against human gold labels.
- **The Ensemble Fix:** Utilizing active token sampling ($T > 0$) and aggregating the 3 runs via a simple **Majority Vote** yields the highest overall accuracy.
- **The Takeaway:** Token variance is actually mathematically necessary for optimal alignment. Consensus pooling is superior to forced determinism.

Methodological Strengths & Insights

- **Rigorous Formulation:** Successfully shifts the automated evaluation literature from raw alignment heuristics to standard chance-corrected reliability frameworks.
- **Multi-Dimensional Granularity:** Uncovers critical insight into metric-specific evaluation performance, revealing that structural model alignment varies drastically by attribute type.
- **Practical Prompt Invariance:** Systematically demonstrates that simple prompt interventions like Chain-of-Thought or Few-Shot examples fail to mitigate underlying self-reliability issues.
- **Avenue Identification:** Exposes major hidden variance components in widely relied upon evaluation leaderboards.

Open Vulnerabilities & Practical Limitations

- **Human Baseline Deficit:** The existing study is structurally unable to establish a human self-reliability upper bound due to complete omission of multi-run annotation data in legacy benchmarks.
- **Reasoning Path Omission:** The analysis bypasses evaluation of internal reasoning pathways or token log probabilities for explicit generation drift tracking.
- **Closed-Model Scope Restrictions:** The experimental regime relies entirely on open-weight models, leaving the stability profiles of massive proprietary APIs unmeasured.
- **Subjectivity Confounding:** Does not decouple inherent human baseline task ambiguity from algorithmic evaluation failure modes.

Strategic Recommendations for Automated Evaluation Pipelines

- **Mandatory Multi-Run Reporting:** Stop publishing single-run LLM evaluation scores; reporting intra-rater reliability statistics must become standard procedure.
- **Deploy Sampling-Based Consensus Pools:** Replace deterministic single-inference workflows with sampled multi-run pools paired with majority-vote aggregation.
- **Establish Multi-Annotator Human Benchmarks:** Structure human data gathering pipelines to include repeated ratings of the same item to identify valid performance upper bounds.
- **Prioritize Chance-Corrected Architecture:** Favor chance-corrected evaluation frameworks like Krippendorff's Alpha over crude percentage overlap metrics.